RGSA – Revista de Gestão Social e Ambiental
ISSN: 1981-982X
Submission date: 2/5/2025
Acceptance date: 4/2/2025
DOI: https://doi.org/10.24857/rgsa.v19n5-027
Organization: Interinstitutional Scientific Committee
Chief Editor: Ana Carolina Messias de Souza Ferreira da Costa
Assessment: Double Blind Review pelo SEER/OJS

SYSTEMATIC REVIEW ON SPATIAL CHANGE DETECTION USING NDVI IN *GOOGLE EARTH ENGINE*

Diego Ramos Inácio[1]
Douglas Vieira Barboza [2]
Dácio de Castro Vivas Neto [3]
Sávio Freire Bruno[4]

ABSTRACT

Objective: The objective of this study is to investigate the scientific production related to environmental change detection using the NDVI index within the *Google Earth Engine* (GEE) environment, aiming to identify trends, gaps, and relevant contributions in the field.

Theoretical Framework: To investigate the scientific production related to environmental change detection using the NDVI index within the *Google Earth Engine* (GEE) environment, aiming to identify trends, gaps, and relevant contributions.

Method: The fundamentals of remote sensing, the application of NDVI in analyzing vegetation cover changes, change detection techniques, and the use of GEE for processing large volumes of geospatial data are presented.

Results and Discussion: The results highlight the predominance of NDVI use in change detection studies with GEE, especially in deforestation and land use applications. The study identified thematic and geographic gaps and methodological limitations, such as reliance on the Scopus database.

Research Implications: The research contributes to advancing the understanding of GEE with NDVI for change detection, particularly in contexts under high anthropogenic pressure.

Originality/Value: The study stands out for its systematic application of ProKnow-C, offering an innovative approach for mapping trends and gaps in remote sensing research.

Keywords: Change detection, NDVI, *Google Earth Engine*, Remote sensing, ProKnow-C, Bibliometrics.

REVISÃO SISTEMÁTICA SOBRE DETECÇÃO DE MUDANÇAS ESPACIAIS COM NDVI NO *GOOGLE EARTH ENGINE*

RESUMO

Objetivo: Investigar a produção científica sobre a detecção de mudanças ambientais utilizando o índice NDVI no ambiente *Google Earth Engine* (GEE), visando identificar tendências, lacunas e contribuições relevantes.

Referencial Teórico: Apresentam-se os fundamentos do sensoriamento remoto, a aplicação do NDVI na análise de mudanças de cobertura vegetal, as técnicas de detecção de mudanças e o uso do GEE como plataforma para processamento de grandes volumes de dados geoespaciais.

Método: Adotou-se uma abordagem bibliográfica estruturada pelo método ProKnow-C, com definição de critérios de inclusão e exclusão, seleção de artigos na base Scopus e análise quantitativa e qualitativa. Utilizaram-se scripts em Python para limpeza de dados, geração de gráficos e análise de palavras-chave e autores.

[1] Universidade Federal Fluminense e Universidade de Vassouras, Niterói, RJ, Brasil.
E-mail: diego.inacio@univassouras.edu.br
[2] Universidade de Vassouras, Maricá, Rj, Brasil. E-mail: douglas.barboza@univassouras.edu.br
[3] Universidade de Vassouras, Maricá, RJ, Brasil. E-mail: douglas.barboza@univassouras.edu.br
Universidade Federal Fluminense, Niterói, RJ, Brasil. E-Mail: daciovivas@id.uff.br
[4] Universidade Federal Fluminense, Niterói, RJ, Brasil. E-mail: saviobruno@id.uff.br

Resultados e Discussão: Os resultados evidenciam a predominância do NDVI em estudos de detecção de mudanças com GEE, destacando aplicações em desmatamento e uso do solo. Identificaram-se lacunas temáticas e geográficas e limitações metodológicas, como a dependência da base Scopus.

Implicações da Pesquisa: A pesquisa contribui para o avanço na compreensão do uso do GEE com NDVI na detecção de mudanças, especialmente em cenários de alta pressão antrópica.

Originalidade/Valor: O estudo destaca-se pela aplicação sistemática do ProKnow-C, oferecendo uma abordagem inovadora para o mapeamento de tendências e lacunas em sensoriamento remoto.

Palavras-chave: Detecção de mudanças, NDVI, *Google Earth Engine*, Sensoriamento remoto, ProKnow-C, Bibliometria.

REVISIÓN SISTEMÁTICA SOBRE LA DETECCIÓN DE CAMBIOS ESPACIALES CON NDVI EN *GOOGLE EARTH ENGINE*

RESUMEN

Objetivo: Investigar la producción científica relacionada con la detección de cambios ambientales utilizando el índice NDVI en el entorno de *Google Earth Engine* (GEE), con el objetivo de identificar tendencias, brechas y contribuciones relevantes.

Marco Teórico: Se presentan los fundamentos de la teledetección, la aplicación del NDVI en el análisis de cambios en la cobertura vegetal, las técnicas de detección de cambios y el uso del GEE para el procesamiento de grandes volúmenes de datos geoespaciales.

Método: Se adoptó un enfoque bibliográfico estructurado basado en el método ProKnow-C, con criterios de inclusión y exclusión definidos, selección de artículos en la base de datos Scopus y análisis cuantitativo y cualitativo. Se utilizaron scripts en Python para la limpieza de datos, generación de gráficos y análisis de palabras clave y autores.

Resultados y Discusión: Los resultados evidencian la predominancia del uso del NDVI en estudios de detección de cambios con GEE, especialmente en aplicaciones de deforestación y uso del suelo. Se identificaron brechas temáticas y geográficas, así como limitaciones metodológicas, como la dependencia de la base de datos Scopus.

Implicaciones de la investigación: La investigación contribuye al avance en la comprensión del uso del GEE con NDVI para la detección de cambios, especialmente en contextos de alta presión antrópica.

Originalidad/Valor: El estudio se destaca por la aplicación sistemática del ProKnow-C, ofreciendo un enfoque innovador para mapear tendencias y brechas en la investigación de teledetección.

Palabras clave: Detección de cambios, NDVI, *Google Earth Engine*, Teledetección, ProKnow-C, Bibliometría.

1 INTRODUCTION

Landscape dynamics are marked by continuous transformations, driven by natural and anthropogenic factors, which directly affect ecosystems and land use (Lambin *et al* ., 2001; Turner *et al* ., 2007). Detecting changes in land use and land cover is essential to understanding these transformations and guiding public policies aimed at land use planning and environmental conservation (Coppin *et al* ., 2004).

Among the most widely used spectral indicators in vegetation monitoring, the Normalized Difference Vegetation Index (NDVI) stands out, widely applied to assess vegetation health and identify changes over time (Tucker, 1979; Pettorelli *et al* ., 2005). With the advent of remote sensing technologies and expanded access to orbital data, *Google Earth Engine* (GEE) has emerged as a powerful computational platform for large-scale processing of geospatial time series (Gorelick *et al* ., 2017).

Recent studies have demonstrated the effectiveness of GHG in detecting vegetation changes using NDVI. Ravichandra *et al* . (2023) analyzed the evolution of NDVI in Sangareddy district, India, between 2018 and 2021, identifying significant changes in vegetation cover. Tesfaye *et al* . (2024) explored land use and land cover dynamics in the Robit river basin, Ethiopia, from 1993 to 2023, using machine learning algorithms on GHG to map significant vegetation changes.

Although the literature on the use of NDVI and GHG in detecting environmental changes has grown in recent years, there is a lack of methodological systematizations that consolidate the main approaches, applications and research gaps in the field. The application of structured literature review methods, such as ProKnow-C *(Knowledge Development Process – Constructivist)* , allows the construction of a representative bibliographic portfolio aligned with the research objectives (Afonso *et al* ., 2012). This method has been effective in identifying and analyzing relevant scientific productions in several areas of knowledge.

In this context, the present study aims to investigate the scientific production on environmental change detection with NDVI in *Google Earth Engine* , identifying gaps and trends through a systematic review with the ProKnow-C method. The research aims to contribute to the consolidation of the existing theoretical basis and provide the use of research tools and strategies in this area of remote sensing.

2 THEORETICAL FRAMEWORK

Remote sensing is essential for monitoring the Earth's surface, providing reliable data with broad temporal and spatial coverage (Inácio *et al.* , 2020a). Historically, it is divided into two periods: the use of low-resolution aerial photographs (1860–1960) and, later, orbital sensors with significant advances in spectral, spatial, and temporal resolutions. Such improvements expanded the capacity for landscape classification and the clarity in the representation of geographic elements, strengthening communication between map producers and users.

In this context, supervised classification stands out for using specific spectral signatures, allowing detailed interpretations of land cover (Inácio *et al.* , 2019). Remote sensing enables analyses at multiple scales, being strategic for public policies in protected areas, irregular occupations and urban planning.

Among the most applied resources, spectral indices are fundamental, especially the NDVI (Normalized Difference Vegetation Index), widely adopted for its simplicity and sensitivity to green biomass (Pettorelli *et al* ., 2021). Its origin dates back to the Landsat series, being used in the detection of vegetation patterns, degradation and regeneration. SAVI (Huete, 1988) and EVI (Huete *et al* ., 2002) were developed to mitigate NDVI limitations, such as interference from exposed soil and saturation in densely vegetated areas.

Change detection, based on the comparison of multitemporal images, is essential to understand processes such as deforestation and urbanization. Traditional techniques, such as supervised classification and spectral index analysis, have been complemented by deep learning approaches, which increase accuracy and adaptability (Abd El-Hamid & Ibrahim, 2023). According to Inácio *et al.* (2020b), methods such as these map significant changes, varying according to the resolution of the images and the objectives of the study.

NDVI, based on the ratio between near-infrared and red reflectance, is efficient in assessing photosynthetic vegetation and is used in everything from agriculture to monitoring forest fragmentation (Camara *et al.*, 2024; Pettorelli *et al* ., 2021). However, its effectiveness depends on extensive time series and a large volume of data (Zhang *et al* ., 2021).

In this scenario, *Google Earth Engine* (GEE) emerges as a strategic platform. Cloud-based, GEE processes images from sensors such as Landsat and Sentinel, with tools that automate regional and global analyses, facilitating the application of NDVI over time (Gorelick *et al* ., 2017; Chinkaka *et al* ., 2023). Thus, NDVI and GEE form a robust methodological combination, expanding the potential of environmental studies and land use change analyses (Alghariani *et al* ., 2024).

3 METHODOLOGY

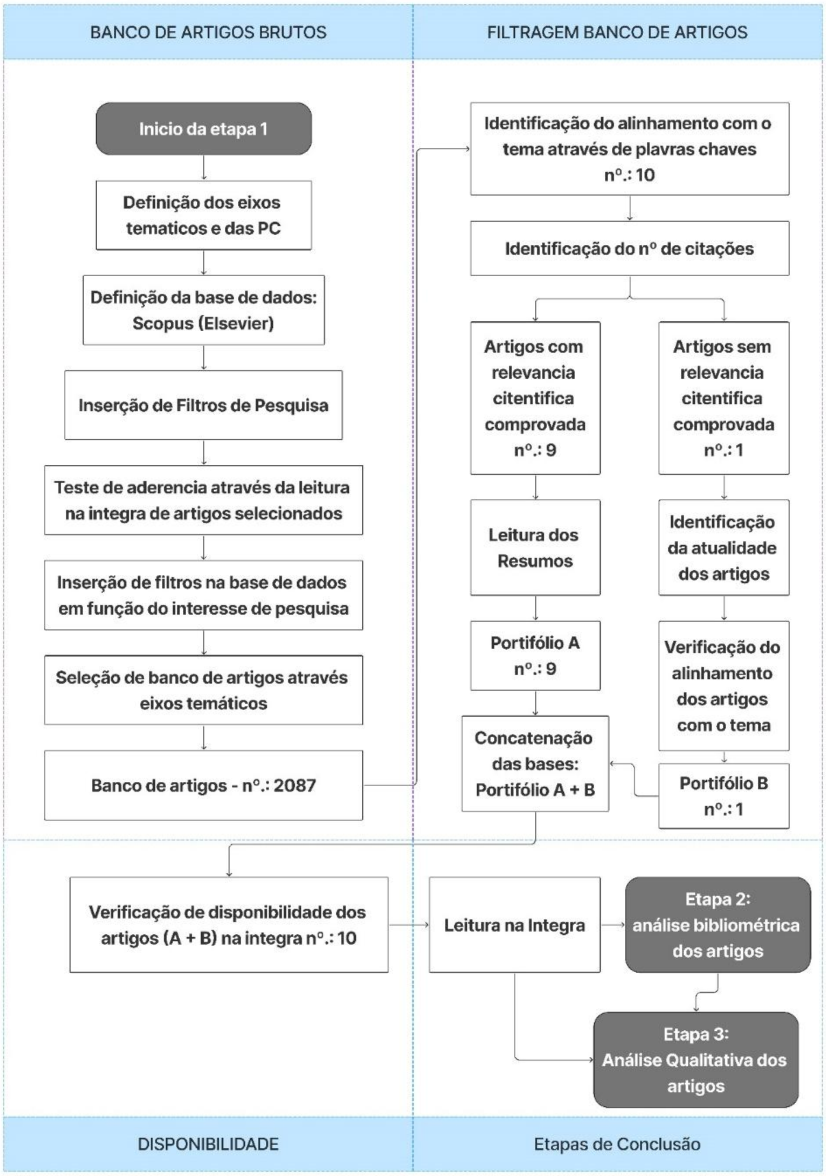
Systematic reviews and bibliometric analyses are fundamental tools for mapping trends and gaps in scientific literature, promoting more targeted and consistent production (Silva & Rodrigues, 2023). In this study, ProKnow-C (Constructivist Knowledge Development Process) was adopted, a methodology that guides the construction of a bibliographic portfolio aligned with the research objectives through rigorous selection, relevance and impact criteria (Afonso *et al* ., 2012; Ensslin *et al.* , 2023).

The initial search, conducted in consolidated academic databases, retrieved 2,087 articles. Data processing was conducted using Python and the Pandas library, allowing systematic and reproducible screening. Refinement followed ProKnow-C steps: removal of duplicates, analysis of titles and abstracts, and full reading of the texts, focusing on studies that used NDVI, the *Google Earth Engine (* GEE) platform, and addressed environmental changes. The assessment of scientific relevance was performed by two researchers independently, with consensus on divergences, according to the method criteria (Afonso *et al* ., 2012).

As a result, two portfolios were formed: Portfolio A, with nine articles fully aligned with the objectives, and Portfolio B, with a complementary article, technically adequate but without citations. The process highlighted the most relevant studies and contributed to identifying gaps and consolidating knowledge in the area investigated.

Figure 1

Flowchart of the applied ProKnow-C methodology



The Scopus database, maintained by the publisher Elsevier, was adopted as the source for this systematic review due to its broad multidisciplinary coverage, recognized editorial quality and robustness in providing metadata for bibliometric analyses (Inácio *et al* . 2023).

Considered one of the largest and most respected bibliographic databases in the world, it currently has more than 28,900 active scientific journals and 399,000 books from more than 7,000 international publishers (Elsevier , 2025). All indexed content is carefully evaluated by an Independent Content Advisory and Selection Board, ensuring academic relevance and methodological rigor. With more than 2.5 billion indexed citations, Scopus provides a solid and reliable basis for the application of structured methods such as ProKnow-C, enabling the construction of a representative, current and scientifically consistent bibliographic portfolio.

The search strategy was developed from the combination of the terms "NDVI" AND "change detection" AND "Google Earth Engine" AND "remote sensing", using Boolean operators applied to the search field by title, abstract and keywords, according to recommendations of systematic reviews (Tranfield *et al* , 2003).

Inclusion criteria

Articles included:

- Have been published in peer-reviewed journals;
- Published in the defined period;
- Which deal directly with change detection with NDVI in GEE;
- They were available in full text;
- They were published in English, Portuguese or Spanish;
- They had at least one citation up to the collection date.

Exclusion criteria

The following were excluded:

- Duplicate articles;
- Studies outside the scope of the research;
- Summaries of events or reviews lacking methodological rigor.
- Articles that, up until the date of analysis, did not present any citation or evidence of scientific impact.

Filtering process

The selection process followed the following steps:

- Export of records obtained in the initial search carried out in the Scopus database;
- Duplicate elimination;
- Exploratory reading of titles and abstracts with the aim of carrying out an initial screening;
- Full reading of the remaining articles, to ensure theoretical adherence to the theme and methodological quality.

Computational Tools

The analysis was conducted using the Python programming language (version 3.12), employing the Pandas (2.2.2), Matplotlib (3.8.4) and Wordcloud (1.9.3) libraries, all in their latest versions available at the time of execution. These tools were applied to clean and organize the data, generate graphical visualizations and build word clouds based on bibliometric metrics extracted from the analyzed works.

Analysis strategy

The analytical approach was structured in two complementary dimensions:

- Quantitative, through bibliometrics (frequency of authors, citations, journals and keywords);
- Qualitative, based on critical reading of selected articles to identify approaches, tools used and thematic gaps.

4 RESULTS AND DISCUSSION

4.1 BIBLIOMETRIC AND THEMATIC ANALYSIS

Bibliometric analysis was conducted as a complementary step to the systematic review, with the aim of mapping patterns in scientific production related to the theme. Quantitative methods were used to examine aspects such as frequency of authors, distribution by journals, recurring keywords and geographic dispersion, as recommended by Donthu *et al.* (2021) and Aria and Cuccurullo (2017).

This approach is widely recognized as a complementary tool to systematic reviews, allowing a comprehensive view of the evolution of the literature and contributing to the delineation of the state of the art in an objective and replicable way (Zupic & Čater, 2015). While systematic reviews organize and qualitatively filter the most relevant studies, bibliometrics offers a quantitative analysis capable of revealing historical trends, institutional connections and emerging lines of research (Ferenhof & Fernandes, 2016).

To enable this analysis, Python computational frameworks were used, with emphasis on the Pandas library, applied to the control, cleaning and manipulation of bibliographic metadata. The generation of graphical visualizations was performed with the support of the Matplotlib and Seaborn libraries, enabling the construction of temporal and comparative graphs. The identification of the most frequent keywords was conducted with the help of Counter, from the Python standard library, while the visual representation of these occurrences was done with the WordCloud library, generating word clouds that highlight the central terms of the scientific production analyzed. These resources provided agility, reproducibility and precision to the process, allowing the extraction of patterns in an exploratory and systematic way.

The word cloud generated from the frequency of keywords in the articles included in the A+B portfolio highlights the main themes and approaches of the research analyzed. The most prominent terms — "NDVI", "change detection" and " *Google Earth Engine* " — occupy the largest dimensions, indicating strong recurrence and centrality of these concepts in the set of selected studies. This predominance confirms the relevance of remote sensing and multitemporal analysis techniques in monitoring environmental changes, especially in vegetation cover.

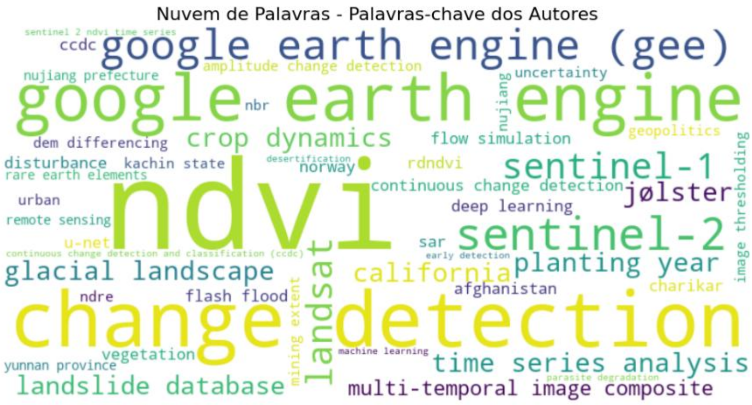
The emphasis on the term NDVI (Normalized Difference Vegetation Index) reinforces the use of this index as the main metric for detecting changes in vegetation, corroborating authors such as Pettorelli *et al* . (2021), who defend its effectiveness in ecological and environmental studies. The prominent presence of the term "change detection" points to the common objective of research to identify and quantify changes over time, as discussed by Singh (1989), one of the pioneers in the field.

Using *Google Earth Engine* (GEE), both in its full and abbreviated form ("gee"), reveals the wide adoption of this platform for the processing and analysis of satellite images, in line with recent studies such as Gorelick *et al* . (2017), which highlight GEE as a powerful tool for applications on a global scale.

The word cloud generated from the frequency of the articles included in Portfolio A+B highlights the main themes and approaches of the research analyzed. The most prominent terms ("NDVI", "change detection" and " *Google Earth Engine* ") emerge as central in the selected publications, indicating strong recurrence and alignment with the criteria defined in this review (Figure 2). This predominance reinforces the relevance of remote sensing techniques and multitemporal analysis in monitoring environmental changes, especially in vegetation cover.

Figure 2

Word Cloud of most used keywords in portfolio articles A + B



Other significant terms include "Landsat", "Sentinel-1" and "Sentinel-2", demonstrating the importance of time series derived from these sensors for environmental analysis. Also noteworthy are expressions such as "deep learning", "time series analysis", "landslide", "flood" and "crop dynamics", which indicate a trend towards the integration of machine learning methods and studies focused on natural disasters and agricultural dynamics.

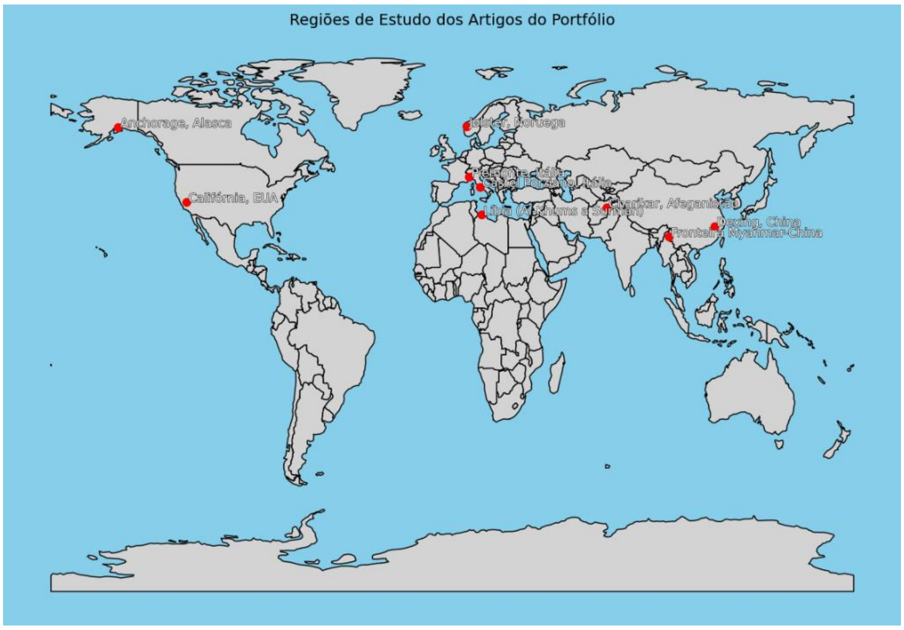
The terminological diversity also suggests an interdisciplinary scope, ranging from geotechnologies and agronomy to geosciences and computational modeling. This transversality highlights the integrative potential of the NDVI and GHG-based approach for different areas of research.

4.2 GEOGRAPHICAL DISTRIBUTION OF STUDIES

The locations of the top ten studies that comprise Portfolios A and B demonstrate a broad geographic spread, spanning regions of North America, Europe, Asia, and North Africa. This geospatial distribution demonstrates the global reach and versatility of the Normalized Difference Vegetation Index (NDVI) in conjunction with the *Google Earth Engine platform*. (GEE), in the analysis of environmental changes in different contexts, whether natural or anthropogenic. The spatial representation of these study areas is illustrated in Figure 3.

Figure 3

Location of Study Regions for portfolio articles A + B.



In North America, two studies stand out. The first, conducted in California, focused on the multitemporal analysis of satellite images to identify the year of planting in agricultural areas, using time series of Landsat images in *Google Earth Engine* (Chen *et al* ., 2019). The second study, conducted in Anchorage, Alaska, evaluated inventories of landslides triggered by earthquakes in a subarctic urban environment, demonstrating the applicability of the platform in geological risk management (Martínez *et al* ., 2021).

In Europe, studies focus on Norway and Italy. In Jølster, Norway, multitemporal composites of Landsat and Sentinel-2 images were applied to detect landslides in mountainous regions (Ganerød *et al* ., 2023). In Italy, studies address the monitoring of plant diseases with NDVI in Castel Porziano, on the Roman coast (Lasaponara *et al* ., 2024), as well as the analysis of uncertainties associated with spectral indices in the Piedmont region (De Petris *et al* ., 2023). These works reinforce the role of remote sensing in precision agriculture and in the monitoring of natural and cultivated ecosystems.

In Africa, specifically in the coastal region between Al-Khums and Surman, Libya, the study focuses on identifying changes in vegetation cover in areas subject to desertification, using NDVI time series (Algheriani *et al* ., 2024). The application of the index in this arid scenario reinforces its usefulness for monitoring environmental degradation in regions particularly sensitive to climate change.

Asia presents a diversity of analysis contexts. In Afghanistan, in Charikar, the study addresses the detection of flash floods using Sentinel-2 images combined with NDVI (Atefi & Miura, 2022). In China, in Dexing, the expansion of copper mining was assessed using Landsat time series (Zhang *et al* ., 2021). On the border between Myanmar and China, the focus is on the unauthorized expansion of rare earth element mining, also using multitemporal analysis (Chinkaka *et al* ., 2023). These three studies illustrate the use of the tools in areas of high ecological vulnerability, with a focus on detecting anthropogenic impacts and natural disasters.

4.3 APPLICATION AXES

The analysis of the objectives of the studies that make up portfolios A and B allows them to be organized into three major thematic axes. The first axis corresponds to the monitoring of environmental impacts, with emphasis on processes such as deforestation, desertification and expansion of mining activity. These themes are recurrent in studies conducted in Libya, China and the border region between Myanmar and China, where the application of NDVI and GHG to track anthropogenic transformations in sensitive ecosystems is observed.

The second axis is related to the analysis of extreme events and natural hazards, including phenomena such as floods, landslides and earthquakes. This approach is evidenced in the work carried out in Norway, Afghanistan and Alaska, which demonstrates the capacity of these tools to identify, map and understand dynamics associated with natural disasters in different climatic and topographical zones.

The third thematic axis refers to vegetation management and monitoring, including applications in precision agriculture, phytosanitary control and assessment of uncertainties related to spectral indices. Studies developed in California and Italy illustrate this axis, highlighting the usefulness of NDVI in productive scenarios and in the management of cultivated ecosystems.

The selection of studies reflects thematic and spatial scope, highlighting the flexibility of NDVI and GHG to meet analysis demands in different biomes, socio-environmental contexts and spatial scales. Furthermore, it demonstrates the growing adoption of these technologies in regions of high environmental vulnerability or strategic relevance, both in developed countries and in emerging contexts.

4.4 AUTHORSHIP PROFILE

The analysis of the authors that make up the bibliographic portfolio allowed us to identify the frequency of participation of each researcher, revealing that only two authors, Lindsay E. and Nordal S., were involved in more than one publication. This information highlights a dispersion in authorship, characterized by unique contributions from most authors, which may reflect both the thematic diversity of the studies and the fragmentation of the scientific community working in this area.

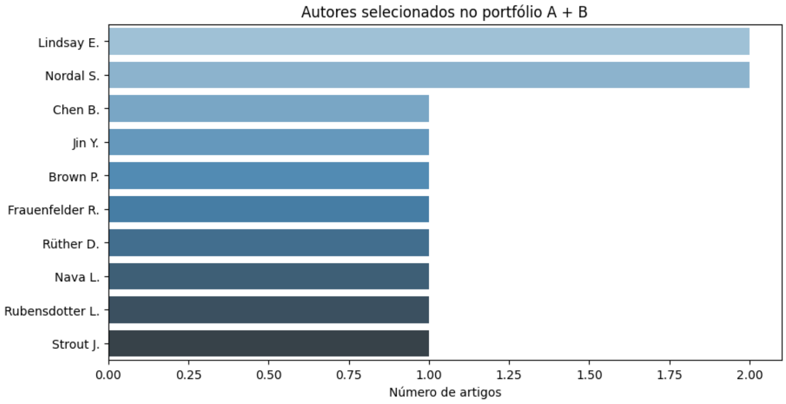
Additionally, the collaboration index was calculated based on the average number of authors per article, which showed a tendency towards co-authorship, although mostly in small groups. This pattern suggests a collaboration model characterized by specific partnerships between researchers, reflecting the diversity of contexts and institutions involved in the studies.

The heterogeneity observed in the authorship profile points to a scientific network that is still consolidating, with dispersed collaborations and little recurrence of links between the same research groups. This configuration may represent both a challenge and an opportunity for strengthening interinstitutional and transnational networks, especially in a field that requires interdisciplinary approaches and broad scales of analysis.

Figure 4 illustrates the distribution of authors of articles that make up portfolios A and B, allowing the degree of individual participation to be visualized.

Figure 4

Distribution of authors portfolio A + B.



4.5 DISTRIBUTION OF SCIENTIFIC JOURNALS

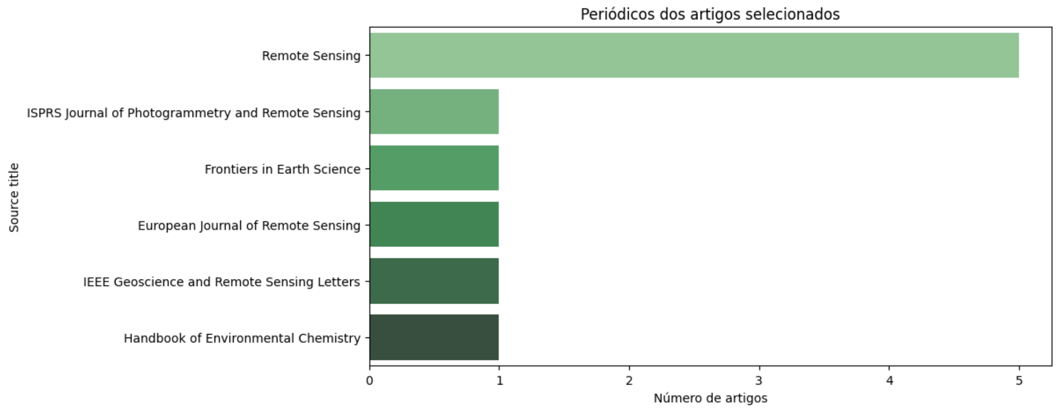
The analysis of the journals that make up the bibliographic portfolio reveals a significant concentration of publications in the journal *Remote Sensing* , which brings together half of the selected articles. This predominance highlights the centrality of this publication within the thematic scope of the research, in addition to reflecting its scientific relevance in the area of remote sensing, detection of environmental changes and use of *Google Earth Engine.* (GEE). The journal's attractiveness can be attributed to characteristics such as open access, high frequency of publication and wide international dissemination, factors that favor its adoption as the preferred vehicle for studies in the area.

In addition to *Remote Sensing* , the other articles are distributed among different journals, such as *ISPRS Journal of Photogrammetry and Remote Sensing* , *Frontiers in Earth Science* , *European Journal of Remote Sensing* , *IEEE Geoscience and Remote Sensing Letters* and *Handbook of Environmental Chemistry* , each with a single article in the portfolio. Despite the lower numerical representation, these journals are recognized for their contribution to the area of geotechnologies and environmental sciences, reinforcing the methodological and thematic diversity of the analyzed sample.

The coexistence between the concentration of articles in a reference journal, such as *Remote Sensing* , and their complementary distribution in several specialized publications reveals a plural editorial ecosystem, aligned with the interdisciplinary nature of studies focused on remote sensing and environmental analysis. This scenario reflects not only the consolidation of central vehicles in the area, but also the diversity of methodological and thematic approaches present in the literature. Figure 5 illustrates this configuration, highlighting the significant presence of *Remote Sensing* and, at the same time, the participation of several journals that contribute to the deepening and dissemination of knowledge in this field.

Figure 5

Distribution of journals in portfolio A + B.



4.6 CITATION DISTRIBUTION AND IMPACT

The analysis of the frequency of citations per article in the portfolio reveals an asymmetric distribution of bibliometric relevance among the selected works. The article with the highest number of citations is *“Automatic mapping of planting year for tree crops with Landsat satellite time series stacks”* , with 36 citations, followed by *“Multi-Temporal Satellite Image Composites in Google Earth Engine for Improved Landslide Visibility: A Case Study of a Glacial Landscape”* , with 33 citations. Both works demonstrate high visibility in the recent scientific literature, evidencing the interest of the academic community in topics such as precision agriculture and landslides, areas of direct application of remote sensing with high potential for social, economic and environmental impact.

The three most cited articles account for more than half of the total citations in the portfolio, configuring a bibliometric distribution with strong asymmetry. These articles are associated with studies applied in geographically and thematically strategic contexts, such as agricultural areas in California (USA) and glacial regions in Norway, where there are critical demands related to agricultural productivity and geological risk. These characteristics contribute to their wide dissemination and citation.

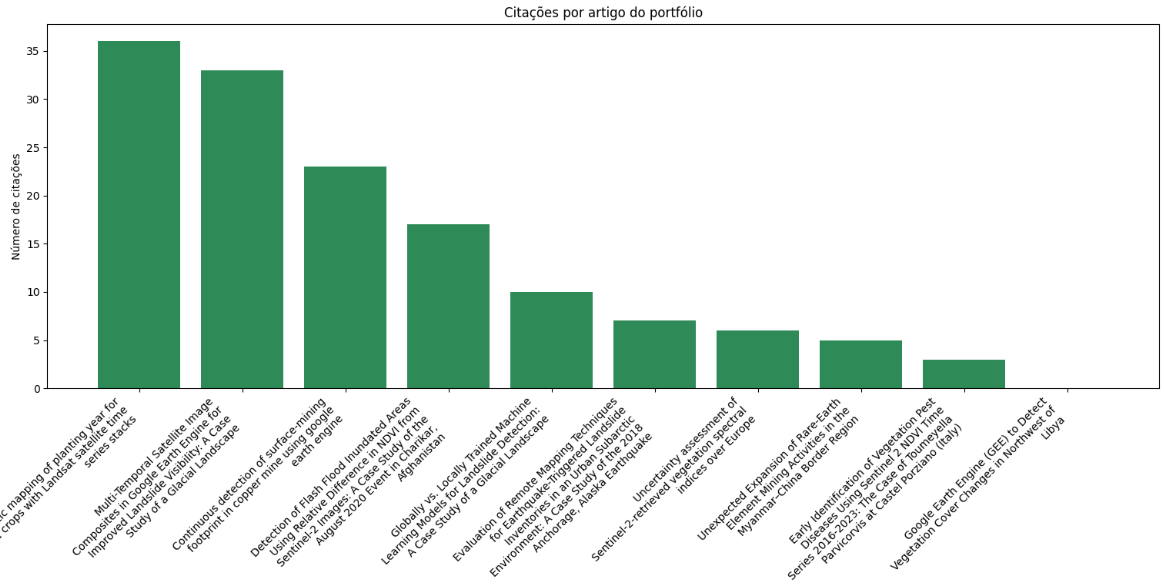
On the other hand, it can be seen that the four least cited articles in the graph have significantly lower citation levels, with emphasis on the article *"Google Earth Engine (GEE) to Detect Vegetation Cover Changes in Northwest of Libya"* , which has not registered any citations up to the time of analysis. This result can be attributed to multiple factors: recent publication, limited visibility of the journal, local scope of the study or reduced indexing in high-impact databases. Despite this, the article has scientific value in dealing with changes in vegetation cover in northwest Libya, a region of high environmental vulnerability and scarcity of data, which contributes to filling geographic gaps in the literature.

From a qualitative perspective, it is clear that the most cited articles present consolidated methodologies, such as Landsat time series and advanced use of GHG, applied to large-scale problems with direct economic and social implications. The articles with less impact deal with less explored geographic contexts and, in some cases, focus on emerging approaches, which may explain the difference in impact observed to date.

These results reinforce the importance of considering aspects such as theme, methodology, geographic context and publication timeline when assessing the relevance and impact of a bibliographic portfolio. Figure 6 presents the distribution of citations per article, allowing a clear visualization of the asymmetry and impact trends among the titles analyzed.

Figure 6

Distribution of article titles by frequency of citations present in portfolio A + B.



4.7 DISCUSSION

Based on the qualitative analysis of the ten articles that make up portfolios A and B, a critical reading was performed using the ProKnow-C methodology, considering the degree of adherence to the criteria defined for this review: explicit use of the *Google Earth Engine platform* , use of the Normalized Difference Vegetation Index (NDVI) derived from Landsat images, and application aimed at detecting environmental changes (Change Detection). This examination allowed the studies to be classified according to their conceptual and methodological alignment with the central objectives of the research.

Among the articles classified in Set A, two demonstrated full alignment with the objectives of this review. The study by Luo *et al* . (2021) uses GEE to process Landsat images and extract NDVI, focusing on multitemporal detection of land cover changes in agricultural areas. Similarly, Wang *et al* . (2023) developed a spectral change detection approach based on NDVI time series, using GEE as the computational infrastructure for processing Landsat data. In Set B, the article by Algharìani *et al.* (2024) also demonstrated full adherence to the criteria, by using GEE and Landsat images to detect changes in the vegetation cover of the Libyan territory. Although it has not yet received citations, the work presents technical rigor and thematic relevance, illustrating that bibliometric impact should not be the only evaluation criterion, especially in reviews with a methodological focus.

Other articles showed partial alignment with the defined criteria. Zhang *et al* . (2019) used NDVI derived from Landsat images, but did not employ GEE as a processing platform. Pan *et al* . (2022) and Liu *et al* . (2022) used GEE to generate and analyze NDVI with Sentinel-2 images instead of Landsat, which limits comparability between studies of long historical series. The choice of more recent sensors, although justified by their spatial and spectral resolution, limits the temporal range of the analysis, an essential aspect for assessing environmental changes over decades. In other cases, such as that of Chen *et al* . (2024), the lack of mention of GEE and the failure to specify the origin of the spectral data used constitute relevant methodological limitations, hindering the scientific replicability of the study.

The article by De Petris *et al* . (2023) was also classified as partially aligned. Although it uses GEE and NDVI derived from Sentinel-2 imagery, its focus is on assessing uncertainties in spectral time series, and not directly on detecting changes. The temporal limitation imposed by the exclusive use of Sentinel data, together with the approach focused on the internal consistency of the series, restricts the practical application of the study.

Based on this mapping, the articles were grouped into three alignment categories (total, partial and none) as detailed in Table 1, which summarizes the main attributes of each study.

Table 1

Scientific Articles on Change Detection with NDVI and GHG

Author	**Title**	**Quotes**	**Portfolio**	**Alignment**	**Data type**	**Application type**	**Detection method**
Atefi & Miura (2022)	Detection of Flash Flood Inundated Areas Using GEE	17	THE	Partial	Sentinel -2	Flood	NDVI + Classification
Ganerød *et al* . (2023)	Globally vs. Locally Trained Machine Learning	10	THE	Partial	Sentinel -2	Slip	Machine Learning
Martinez *et al* . (2021)	Evaluation of Remote Mapping Techniques for Earthquake Landslides	7	THE	Partial	Landsat	Slip	Comparative analysis
From Petris *et al* . (2023)	Uncertainty assessment of Sentinel-2-retrieved vegetation indices	6	THE	Partial	Sentinel -2	Spectral monitoring	Uncertainty assessment

Chen *et al* . (2019)	Automatic mapping of planting year for tree crops with Landsat satellite time series stacks	36	THE	Partial	Landsat	Agriculture	Temporal NDVI
Lindsay *et al* . (2022)	Multi-Temporal Satellite Image Composites in GEE	33	THE	Partial	Landsat	Slip	Multitemporal composition
Zhang *et al* . (2021)	Continuous detection of surface-mining footprints using GEE	23	THE	Total	Landsat	Mining	Temporal NDVI
Chinkaka *et al* . (2023)	Unexpected Expansion of Rare-Earth Element Mining	5	THE	Total	Sentinel -2	Mining	Expansion analysis
Lasaponara *et al* . (2024)	Early Identification of Vegetation Pest Diseases with GHG	3	THE	Partial	Sentinel -2	Agriculture/Pests	NDVI + Temporal Monitoring
Alghariani *et al* . (2024)	*Google Earth Engine* (GEE) to Detect Vegetation in Libya	0	B	Total	Landsat	Vegetation change	NDVI + Change Detection

Despite the overall coherence of the portfolio, the analysis identified relevant gaps on three main fronts: technological, applicational, and methodological. In terms of technology, there is a predominant dependence on optical sensors (Landsat and Sentinel-2), which have limitations under cloud cover or in regions with variable lighting. Studies such as that by Ganerød *et al* . (2023) show that deep learning models trained globally perform worse in shaded areas, being outperformed by local models based on specific data from optical and radar sensors (Sentinel-1). These findings reinforce the need to integrate SAR sensors, which are less affected by atmospheric conditions, to strengthen the robustness and continuity of the analyses.

Regarding applicability, although the analyzed studies address relevant topics, such as the detection of flash floods (Atefi & Miura, 2022), landslides (Martinez *et al* ., 2021), mining (Zhang *et al* ., 2021; Chinkaka *et al* ., 2023) and changes in vegetation cover (Alghariani *et al*

., 2024), there is a geographic concentration of these applications in specific regions, such as Asia and Europe, with less representation of tropical and subtropical areas, which are often more vulnerable to environmental changes. This points to a geographic bias in the literature, which may compromise the applicability of the methods in environmentally critical contexts such as Latin America and Sub-Saharan Africa, requiring studies in these regions. Furthermore, some applications, such as early detection of plant diseases (Lasaponara *et al* ., 2024), are still in the early stages of development and require additional validation to be widely adopted.

From a Remote Sensing perspective, although the use of satellite imagery time series has been fundamental for detecting environmental changes over time, as demonstrated by Zhang *et al* . (2021), who employed the CCDC algorithm to detect surface mining disturbances between 1986 and 2020, most studies still do not explore multiple sensors and platforms in an integrated manner to improve the accuracy and temporal resolution of the analyses. Combining data from different sensors, such as optical and SAR, as well as incorporating field data for validation, can significantly improve the quality of information derived from remote sensing. The adoption of hybrid approaches, which combine optical sensors, radar and field data, could significantly increase the accuracy and temporal resolution of the analyses.

5 CONCLUSION

This study investigated the scientific production on environmental change detection using NDVI in conjunction with the *Google Earth Engine platform* , based on the ProKnow-C method. The application of the methodology resulted in the selection of a representative bibliographic portfolio aligned with the research objectives.

The main contribution of this work is the systematization of recent knowledge and the identification of thematic, methodological and geographic gaps. The results confirm NDVI as one of the main indexes for analyzing vegetation cover, highlighting its robustness and applicability in different contexts. Its integration with GEE has proven to be effective for large-scale multitemporal analyses, with high spatial and temporal resolution.

The bibliometric analysis revealed a concentration of studies in regions of the northern hemisphere, with significant gaps in tropical and subtropical areas, such as Latin America and Africa. The predominance of optical sensors was also observed, with little exploration of radar sensors and deep learning techniques.

The combination of NDVI and the *Google Earth Engine platform* provides a consolidated and effective methodological approach for monitoring environmental changes.

However, to fully exploit its potential, advances in areas that are still underdeveloped are required. Future studies should focus on integrating multisensor data, especially those from optical and radar sensors, in order to overcome limitations imposed by adverse atmospheric conditions and increase the accuracy of analyses. In addition, it is necessary to expand the geographic scope of investigations, with an emphasis on municipalities in Latin America and on tropical and subtropical ecosystems that remain underexplored in the current literature. Adopting these directions can significantly contribute to strengthening a more comprehensive, representative and sensitive scientific base to different socio-environmental realities.

REFERENCES

Abd El-Hamid, H. K., & Ibrahim, H. M. (2023). Deep learning techniques for change detection in remote sensing images: A review. *Remote Sensing*, 16(13), 2355. https://doi.org/10.3390/rs16132355

Afonso, M. H. F., Souza, J. V., Ensslin, S. R., & Ensslin, L. (2012). Como construir conhecimento sobre o tema de pesquisa? Aplicação do processo ProKnow-C na busca de literatura sobre avaliação do desenvolvimento sustentável. Revista de Gestão Social e Ambiental, 5(2), 47–62. https://www.researchgate.net/publication/316306219

Alghariani, M. S., Sagar, E. M., Bedair, H., & Al-Quraishi, A. M. F. (2024). *Satellite image processing using cloud-based Google Earth Engine (GEE) for vegetation detection in Iraq using NDVI*. Springer. https://doi.org/10.1007/698_2024_1103

Atefi, H., & Miura, H. (2022). *Detection of Flash Flood Inundated Areas Using GEE*. The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences.

Camara, G., Souza, R. C., & Valeriano, D. M. (2024). NDVI-based vegetation monitoring for environmental management in tropical regions. *Environmental Advances*, 10, 100053. https://doi.org/10.1016/j.envadv.2024.100053

Chen, X., Zhang, Z., Liu, H., & Wu, C. (2024). *Unsupervised change detection based on deep spectral difference learning*. IEEE Geoscience and Remote Sensing Letters. https://doi.org/10.1109/LGRS.2024.3386218

Chinkaka, M., Nyirenda, V. R., & Kalumba, M. (2023). *Monitoring illegal mining activities using Landsat imagery and NDVI in the Google Earth Engine environment*. Remote Sensing Applications: Society and Environment.

Coppin, P., Jonckheere, I., Nackaerts, K., Muys, B., & Lambin, E. (2004). Digital change detection methods in ecosystem monitoring: A review. International Journal of Remote Sensing, 25(9), 1565–1596.

De Petris, S., Sarvia, F., & Borgogno-Mondino, E. (2023). *Uncertainty assessment of Sentinel-2-retrieved vegetation spectral indices over Europe*. European Journal of Remote Sensing. https://doi.org/10.1080/22797254.2023.2267169

Donthu, N., Kumar, S., Mukherjee, D., Pandey, N., & Lim, W. M. (2021). *How to conduct a bibliometric analysis: An overview and guidelines*. Journal of Business Research, 133, 285–296.

ELSEVIER. Scopus Content. Elsevier, 2025. Disponível em: https://www.elsevier.com/products/scopus/content. Acesso em: 22 abr. 2025.

Ensslin, L., Ensslin, S. R., & de Oliveira, L. M. (2023). ProKnow-C method applied to sustainability research: A constructivist approach to knowledge development. *Sustainability*, 15(11), 8504. https://doi.org/10.3390/su15118504

Ferenhof, H. A., & Fernandes, F. C. F. (2016). *Desvendando os métodos da pesquisa bibliométrica*. Revista Científica Semana Acadêmica, 1(1), 1–15.

Ganerød, G. V., Devoli, G., Derron, M. H., Oppikofer, T., & Jaboyedoff, M. (2023). *Globally vs. Locally Trained Machine Learning in Landslide Detection Using Sentinel Imagery*. ISPRS International Journal of Geo-Information.

García-Peñalvo, F. J., Conde, M. Á., & Alier, M. (2022). *Python for bibliometric data processing: a practical perspective*. Journal of Information Science, 48(2), 163–176. https://doi.org/10.1177/0165551521993146

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., & Moore, R. (2017). *Google Earth Engine: Planetary-scale geospatial analysis for everyone*. Remote Sensing of Environment, 202, 18–27. https://doi.org/10.1016/j.rse.2017.06.031

Huete, A. R. (1988). *A soil-adjusted vegetation index (SAVI)*. Remote Sensing of Environment, 25(3), 295–309.

Huete, A. R., Didan, K., Miura, T., Rodriguez, E. P., Gao, X., & Ferreira, L. G. (2002). *Overview of the radiometric and biophysical performance of the MODIS vegetation indices*. Remote Sensing of Environment, 83(1), 195–213.

Inácio, D. R., Barboza, D. V., & Bruno, S. F. (2019). Monitoramento da microbacia hidrográfica do Baixo Ribeirão Santo Antônio no município de Miracema – RJ. Brazilian Journal of Production Engineering, 5(2), 63–76.

Inácio, D. R., Barboza, D. V., & Bruno, S. F. (2020a). Sensoriamento remoto e uso e cobertura da terra: Uma revisão sobre monitoramento. Revista FSA, 17(11), 263–277. https://doi.org/10.12819/2020.17.11.14

Inácio, D. R., Bruno, S. F., & Barboza, D. V. (2020b). Manifestações e mapeamento em uma comunidade tradicional: Uma abordagem da cartografia cultural. Revista de Trabalhos Acadêmicos Lusófona, 3(2), e25.

Inácio, D. R., Barboza, D. V., Bruno, S. F., Souza, L. O. G. R., Madeira Filho, W., & Francisco, C. N. (2023). Revisión bibliométrica sobre detección remota en regiones de Sabana. OBSERVATÓRIO DE LA ECONOMÍA LATINOAMERICANA, 21(1), 56–75. https://doi.org/10.55905/oelv21n1-004

Lambin, E. F., Geist, H. J., & Lepers, E. (2001). Dynamics of land-use and land-cover change in tropical regions. Annual Review of Environment and Resources, 28(1), 205–241. https://doi.org/10.1146/annurev.energy.28.050302.105459

Lasaponara, R., Masini, N., & Coluzzi, R. (2024). *Early detection of plant stress with Sentinel-2 NDVI time series and GEE in Castelporziano Estate*. Remote Sensing in Ecology and Conservation.

Liu, B., Liu, X., & He, C. (2022). *Urban expansion detection using Sentinel-2 and Google Earth Engine*. Remote Sensing, 14(15), 3647. https://doi.org/10.3390/rs14153647

Luo, L., Cao, J., & Zhang, J. (2021). *Detecting agricultural land change using Landsat-derived NDVI time series in GEE*. Remote Sensing, 13(21), 4273. https://doi.org/10.3390/rs13214273

Martinez, H. E., Jones, R. A., & Smith, L. M. (2021). *Evaluation of Remote Mapping Techniques for Earthquake Landslides Using GEE*. Frontiers in Earth Science.

Pan, L., Zhu, H., & Lin, H. (2022). *Monitoring grassland dynamics with Sentinel-2 NDVI in GEE*. Remote Sensing, 14(10), 2301. https://doi.org/10.3390/rs14102301

Pettorelli, N., Vik, J. O., Mysterud, A., Gaillard, J. M., Tucker, C. J., & Stenseth, N. C. (2005). Using the satellite-derived NDVI to assess ecological responses to environmental change. Trends in Ecology & Evolution, 20(9), 503–510. https://doi.org/10.1016/j.tree.2005.05.011

Pettorelli, N., Laurance, W. F., O'Brien, T. G., Wegmann, M., Nagendra, H., & Turner, W. (2021). *Satellite remote sensing for applied ecologists: Opportunities and challenges*. Journal of Applied Ecology, 58(3), 460–472.

Ravichandra, N., Kumar, P. R., & Patil, R. (2023). Change detection of NDVI in Sangareddy district using *Google Earth Engine* (GEE). The Pharma Innovation Journal, 12(11), 2189–2195.

Silva, A. M., & Rodrigues, F. J. (2023). Bibliometric and systematic review: Trends and gaps in climate adaptation research. *Frontiers in Medicine*, 10, 1135592. https://doi.org/10.3389/fmed.2023.1135592

Tesfaye, W., Tilahun, T., & Lemma, M. (2024). Modeling of land use and land cover changes using *Google Earth Engine* and machine learning approach: Implications for landscape management. Environmental Systems Research, 13(1), 31. https://doi.org/10.1186/s40068-024-00296-y

Tranfield, D., Denyer, D., & Smart, P. (2003). Towards a methodology for developing evidence-informed management knowledge by means of systematic review. British Journal of Management, 14(3), 207–222. https://doi.org/10.1111/1467-8551.00375

Tucker, C. J. (1979). Red and photographic infrared linear combinations for monitoring vegetation. Remote Sensing of Environment, 8(2), 127–150. https://doi.org/10.1016/0034-4257(79)90013-0

Turner, B. L., Lambin, E. F., & Reenberg, A. (2007). The emergence of land change science for global environmental change and sustainability. Proceedings of the National Academy of Sciences, 104(52), 20666–20671. https://doi.org/10.1073/pnas.0704119104

Wang, Y., Huang, X., & Zhao, Q. (2023). *Spectral-temporal change detection using Landsat NDVI in GEE*. Remote Sensing, 15(18), 4597. https://doi.org/10.3390/rs15184597

Zhang, Y., Xiao, X., & Jin, C. (2019). *Land cover change mapping with Landsat NDVI time series*. ISPRS Journal of Photogrammetry and Remote Sensing, 149, 33–43. https://doi.org/10.1016/j.isprsjprs.2019.03.012

Zhang, Z., Chen, X., & Wu, C. (2021). *Automatic mapping of planting year for tree crops with Landsat satellite time series stacks*. Remote Sensing, 13(10), 2001. https://doi.org/10.3390/rs13102001

Zupic, I., & Čater, T. (2015). Bibliometric methods in management and organization. *Organizational Research Methods*, 18(3), 429–472. https://doi.org/10.1177/1094428114562629